A Project Report on

Explainable Healthcare Cost Prediction

Submitted in partial fulfillment for the award of

Bachelor of Technology

Degree

in

Computer Science and Engineering

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CERTIFICATE

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DECLARATION

We declare that this project work is composed by ourselves, that the work contained herein is our own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

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List of Abbreviations

SPARCS	-	Statewide Planning and Research Cooperative System
BMI	-	Body Mass Index
API	-	Application Programming Interface.
SRS	-	Software Requirement Specification
SHAP	-	SHapley Additive exPlanations

Abstract

The increasing cost of healthcare services has created a critical need for intelligent systems capable of estimating medical expenses in advance. This project presents an explainable machine learning–based approach for predicting healthcare costs using patient demographic and clinical data. The dataset is systematically preprocessed through data cleaning, handling of missing values, outlier treatment, and encoding of categorical variables to ensure data quality and consistency.

Key features such as age, gender, length of stay, severity of illness, and other clinical attributes are utilized to train predictive models. Advanced machine learning algorithms, including HistGradientBoostingRegressor and Decision Tree Regressor, are employed to model complex relationships between patient characteristics and healthcare costs. The model is trained and evaluated using performance metrics such as R^2 score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).

Furthermore, the system incorporates explainability using SHAP (SHapley Additive exPlanations), enabling interpretation of model predictions and identification of key cost-driving factors. The results demonstrate strong predictive performance and provide valuable insights into the variables influencing healthcare expenses.

This system can assist patients in financial planning, support healthcare providers in resource optimization, and enable policymakers to make informed, data-driven decisions, thereby improving transparency and efficiency in the healthcare ecosystem.

1. Introduction

Healthcare cost prediction is a significant application of data science and machine learning that aims to estimate future medical expenses based on patient demographic, clinical, and lifestyle information. In recent years, the cost of healthcare services has increased rapidly, making it challenging for individuals to plan and manage their medical expenditures. This has created a strong need for intelligent systems that can accurately and efficiently predict healthcare costs in advance. The primary objective of healthcare cost prediction is to analyze historical healthcare data and identify patterns that influence medical expenses.

The data used for prediction typically includes attributes such as age, gender, body mass index (BMI), length of stay, severity of illness, risk of mortality, and other clinical and demographic factors. By applying machine learning techniques to this data, the system can learn complex relationships between these variables and the corresponding healthcare costs.

In this project, multiple machine learning algorithms such as Decision Tree Regressor, HistGradientBoostingRegressor, and Lasso Regression are used to build predictive models. These models are trained and evaluated on large-scale healthcare datasets to determine their effectiveness in predicting medical costs. Among these, the HistGradientBoostingRegressor demonstrates superior performance due to its ability to efficiently handle large datasets, capture non-linear relationships, and provide better generalization.

Healthcare cost prediction systems provide valuable benefits to multiple stakeholders. Patients can use these predictions to plan their finances and avoid unexpected medical expenses. Hospitals and healthcare providers can improve resource allocation and optimize treatment planning. Insurance companies can utilize these predictions for premium calculation, risk assessment, and policy design. Overall, such systems contribute to improved transparency, better decision-making, and enhanced efficiency in the healthcare ecosystem.

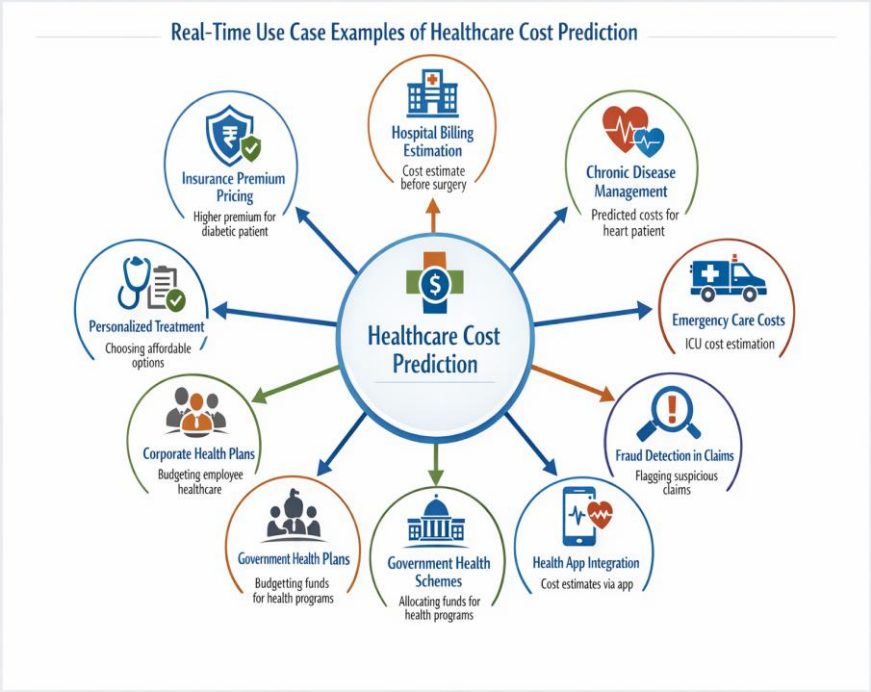

figure 1.1: real-time use case of healthcare cost prediction

figure 1.1: real-time use case of healthcare cost prediction this figure illustrates the real-time applications of the healthcare cost prediction system, where estimated costs support corporate and government health plans, public health schemes, and mobile health app integration. It highlights how predicted costs help in budget allocation, suspicious claim identification, and decision-making across healthcare services.

1.1 Motivation

The motivation behind the Healthcare Cost Prediction project arises from the rapid increase in healthcare expenses and the lack of transparency in medical pricing. Many patients face financial challenges due to uncertainty about treatment costs prior to hospitalization. Additionally, hospitals and insurance providers often follow complex pricing structures, making it difficult for individuals to estimate medical expenses in advance. This highlights the need for an intelligent system capable of predicting healthcare costs based on patient demographic and clinical information.

With the advancement of machine learning techniques, it has become possible to analyze large-scale healthcare datasets and identify patterns that significantly influence medical expenses. By utilizing algorithms such as Decision Tree Regressor, HistGradientBoostingRegressor, and Lasso Regression, the system can learn from

historical patient data and generate accurate cost predictions. These predictions assist patients in financial planning, reduce unexpected expenditures, and support informed decision-making for healthcare providers and insurance companies.

Another important motivation for this project is to improve healthcare efficiency and resource management. Hospitals can leverage cost prediction systems to optimize resource allocation, while insurance companies can estimate claims and determine appropriate premium structures. Furthermore, policymakers and government organizations can use such predictive insights to design better healthcare policies and control rising healthcare costs.

Therefore, the main motivation of this project is to develop an accurate and reliable healthcare cost prediction system that enhances financial planning, increase cost transparency, supports data-driven, decision-making, and contributes to the overall improvement of the healthcare ecosystem.

1.2 Problem Statement

Healthcare costs are increasing rapidly worldwide, making it difficult for patients, hospitals, and insurance providers to manage medical expenses effectively. Patients often lack prior knowledge of treatment costs, leading to financial uncertainty and inadequate planning. Healthcare expenses are influenced by multiple factors such as patient demographics, diagnosis, severity of illness, length of hospital stay, and treatment procedures. Due to the complexity and variability of these factors, accurately predicting healthcare costs has become a challenging problem in modern healthcare systems.

Traditional cost estimation methods are often inefficient and lack accuracy, as they are unable to process large-scale healthcare data and capture complex relationships among variables. With the availability of extensive healthcare datasets, machine learning techniques provide an opportunity to analyze patient information, identify hidden patterns, and build predictive models for cost estimation.

However, most existing approaches focus only on prediction accuracy and do not provide sufficient interpretability. In critical domains such as healthcare,

understanding the factors influencing predictions is equally important for trust and decision-making.

Therefore, there is a need to develop an intelligent and explainable healthcare cost prediction system using advanced machine learning techniques such as Decision Tree Regressor, HistGradientBoostingRegressor, and Lasso Regression. The system should not only predict healthcare expenses accurately but also provide insights into the key factors affecting costs using explainable AI methods such as SHAP.

The proposed solution aims to improve cost transparency, support effective financial planning, and assist hospitals, insurance companies, and policymakers in making data-driven decisions, thereby enhancing the overall efficiency and reliability of healthcare management systems.

1.3 Objective of the Project

The primary objective of this project is to develop an intelligent and reliable healthcare cost prediction system using machine learning and explainable AI techniques. The system is designed to analyze patient demographic and clinical data to accurately estimate future healthcare expenses. By providing early cost predictions, the system enables patients, hospitals, and insurance providers to make informed financial and medical decisions, thereby improving cost transparency and reducing unexpected healthcare expenditures.

Another important objective is to identify and analyze the key factors that significantly influence healthcare costs, such as age, gender, length of hospital stay, diagnosis codes, and severity of illness. The project incorporates explainable AI techniques, specifically SHAP (SHapley Additive Explanations), to provide clear and interpretable insights into how each feature contributes to the predicted cost. This enhances the transparency of the model and builds trust among healthcare professionals and users.

Additionally, the project aims to support efficient healthcare planning and resource management. Accurate cost predictions can assist hospitals in optimizing resource allocation, help insurance companies design better policies and estimate claims, and enable patients to plan their medical expenses effectively. The system

utilizes advanced machine learning models such as Decision Tree Regressor, HistGradientBoostingRegressor, and Lasso Regression to ensure high predictive performance and reliability.

Overall, the objective of this project is to build a robust, accurate, and explainable healthcare cost prediction system that enhances decision-making, improves financial planning, and contributes to the efficiency and transparency of the healthcare system.

2. LITERATURE SURVEY

Healthcare cost prediction has become an important research area due to the rapid increase in medical expenses worldwide. Researchers are using machine learning techniques to predict healthcare costs and help hospitals, insurance companies, and patients make better financial decisions. The paper "Building Predictive Models of Healthcare Costs with Open Healthcare Data" proposes a machine learning-based model to predict healthcare costs using patient demographics and diagnosis information.

The Literature Survey section reviews previous studies and resources used in healthcare cost prediction. These references help understand existing research, identify gaps, and justify the proposed system.

A. Ravishankar Rao, Subrata Garai, Soumyabrata Dey, and Hang Peng proposed a healthcare cost prediction model using machine learning techniques and open healthcare datasets. The main objective of their research was to predict healthcare costs based on patient demographics, diagnosis codes, and severity of illness. The researchers used the New York State SPARCS dataset, which consists of approximately 2.3 million patient records. They applied various machine learning algorithms such as Decision Tree, Linear Regression, Ridge Regression, and Lasso Regression to build predictive models. Among these methods, the Decision Tree model achieved the best performance with an R^2 value of 0.76, indicating high prediction accuracy. The study also identified important features such as patient age, gender, diagnosis code, length of hospital stay, and severity of illness that significantly influence healthcare costs. This research demonstrated that machine learning models can effectively predict healthcare expenses and assist hospitals, insurance companies, and patients in financial planning. However, the study lacked explainability, making it difficult to understand how the model made predictions.

Schwartz et al. conducted a study to identify low-value healthcare services in the Medicare system that increase healthcare costs without providing significant benefits to patients. The researchers analyzed Medicare data, patient records, and treatment costs to identify unnecessary medical procedures. Their findings revealed that several treatments were being performed even though they offered minimal improvement in patient health outcomes. These unnecessary services contributed to

increased healthcare expenses and reduced system efficiency. The study emphasized the importance of reducing unnecessary treatments to improve healthcare quality and minimize costs. Although this research did not develop a predictive model, it highlighted the importance of healthcare cost prediction systems and the need for data-driven decision-making to reduce unnecessary healthcare spending.

The New York State SPARCS (Statewide Planning and Research Cooperative System) dataset is a widely used healthcare dataset provided by the New York State Department of Health. This dataset contains comprehensive healthcare information including patient demographics, diagnosis details, treatment procedures, hospital information, and healthcare costs. Researchers use this dataset to build predictive models for healthcare cost estimation and medical decision-making. The SPARCS dataset is valuable because it includes a large number of patient records and real-world healthcare data, which improves machine learning model accuracy and performance. The availability of detailed healthcare information makes it suitable for training advanced machine learning algorithms. However, since the dataset is limited to New York State, it may not fully represent healthcare systems in other regions or countries.

The Healthcare Price Transparency Initiative was introduced to improve transparency in healthcare pricing and reduce overall medical expenses. This initiative requires hospitals and healthcare providers to publish treatment costs and provide price estimates for medical procedures. The main objective is to help patients compare healthcare costs across hospitals and make informed financial decisions. This policy also encourages competition among healthcare providers, which can lead to reduced medical expenses and improved healthcare efficiency. The initiative supports the development of healthcare cost prediction systems by making cost data publicly available. Although the initiative improves transparency, its adoption is still limited, and data availability remains a challenge in some healthcare institutions.

2.1 Existing System

In the existing system, healthcare cost estimation is mainly done using traditional statistical methods and manual calculations. Hospitals and insurance companies rely on historical data, fixed pricing models, and expert opinions to estimate medical expenses. These methods consider only limited factors such as patient age, diagnosis, and treatment type, which reduces prediction accuracy.

Earlier research used models like multiple regression, linear regression, and classification trees for predicting healthcare costs. However, these models were applied to smaller datasets and produced lower accuracy values. For example, previous studies reported R^2 values such as 0.61, 0.66, and 0.71, showing that traditional models were not highly effective in predicting healthcare costs accurately.

Another limitation of the existing system is that healthcare pricing information is not easily available to patients. Hospitals and insurance companies often keep cost data private, making it difficult for individuals to estimate future medical expenses. Also, traditional systems cannot efficiently handle large healthcare datasets and fail to identify complex relationships between different patient features.

Table 2.1 Existing Models Comparision.

Model Type	RMS Error	R2
Ridge Regression	$13,239	0.71
LassoLarsIC-AIC	$13,190	0.72
LassoLarsIC-BIC	$13,407	0.71
Decision Tree	$12,652	0.76

2.2 Challenges of Existing system

The existing healthcare cost estimation system has several limitations that affect accuracy, transparency, and decision-making. The following are the major disadvantages of the existing system:

- Healthcare cost estimation is mostly done manually, which is time-consuming.
- Traditional methods provide less accurate cost predictions.
- Existing systems cannot handle large healthcare datasets efficiently.
- Lack of price transparency for patients before treatment.
- Difficulty in predicting healthcare costs based on multiple factors.
- Limited use of advanced machine learning techniques.
- Traditional models fail to capture complex relationships in data.
- High chances of human error in manual cost estimation.
- Poor financial planning for patients and hospitals.

- Inefficient resource allocation in healthcare organizations.
- Existing systems do not identify important cost-influencing factors.
- Lack of explainability in cost estimation methods.
- Difficulty in analyzing patient demographic and clinical data together.
- Slow processing of healthcare data.
- Inability to provide real-time cost predictions.
- Limited support for decision-making in hospitals and insurance companies.
- Inconsistent cost estimation across different healthcare providers.
- Existing methods are not scalable for large-scale healthcare systems.
- Lack of automation in healthcare cost prediction.
- Poor utilization of available healthcare data.

2.3 Proposed System

The proposed system introduces a machine learning-based healthcare cost prediction model to accurately estimate future medical expenses. This system uses large healthcare datasets and advanced machine learning algorithms to analyse patient information such as age, gender, diagnosis, severity of illness, length of hospital stay, and other medical factors. By analysing these features, the system predicts healthcare costs more accurately and efficiently.

In the proposed system, data is first collected from healthcare datasets and then preprocessed to remove missing values and irrelevant information. After preprocessing, feature selection techniques are applied to identify important factors affecting healthcare costs. Machine learning algorithms such as HistGradientBoostingRegressor, decision tree, lasso are then used to build prediction models. The system trains the model using training data and evaluates performance using testing data to ensure accuracy.

The proposed system also uses feature importance techniques to understand which factors influence healthcare costs the most. This helps hospitals, insurance companies, and patients make better financial decisions. The system is designed to handle large datasets, improve prediction accuracy, and provide faster results compared to traditional methods.

Overall, the proposed system improves healthcare cost prediction by using advanced machine learning techniques, automated data processing, and efficient model

evaluation. This system helps in financial planning, improves decision-making, and supports healthcare organizations in managing medical expenses effectively.

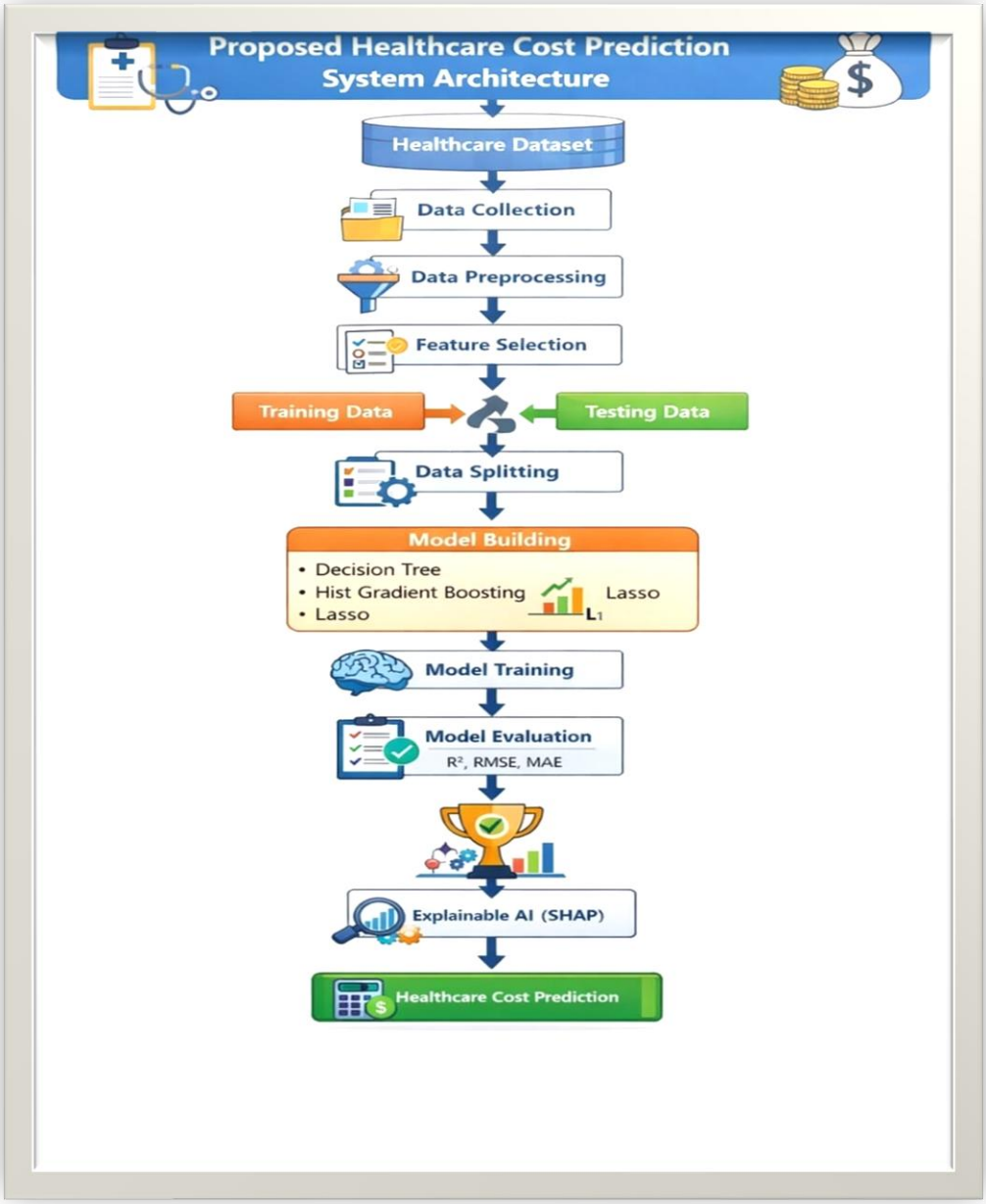

figure 2.1 proposed system architecture

figure 2.1: proposed system architecture this figure presents the workflow of the proposed healthcare cost prediction system, starting from dataset collection, preprocessing, feature selection, and train-test splitting to model building and evaluation. It further includes Explainable AI using SHAP to interpret predictions before generating the final healthcare cost estimation.

Table 2.2 New Models Comparision.

Model Type	RMS Error	R2
Decision Tree	$ 11,835	0.79
Hist Gradient Boosting	$ 10,002	0.85

3. REQUIREMENT SPECIFICATION

System analysis is the process of studying and understanding the existing healthcare cost prediction methods and identifying their limitations. It also involves defining the requirements for developing a more efficient and accurate healthcare cost prediction system. In this project, system analysis focuses on analysing healthcare data, identifying cost-influencing factors, and designing a machine learning-based system to predict healthcare costs effectively.

In the current healthcare system, cost prediction is often performed using traditional statistical methods and manual estimation. These methods are not capable of handling large volumes of healthcare data and complex relationships between patient information and treatment costs. As a result, the predictions are often inaccurate and unreliable. Additionally, the existing system lacks transparency, making it difficult for patients and healthcare providers to understand how costs are determined.

The proposed system addresses these limitations by introducing machine learning and explainable AI techniques. The system collects patient demographic details, diagnosis information, and treatment data, then processes and analyses this data using advanced algorithms such as HistGradientBoostingRegressor, Decision Tree, and Lasso. The system also uses explainable AI methods to identify the most important factors affecting healthcare costs. This improves prediction accuracy, enhances transparency, and supports better decision-making.

Through system analysis, the project identifies the need for an automated, scalable, and intelligent healthcare cost prediction system. The proposed system improves cost estimation, reduces manual effort, increases efficiency, and helps patients, hospitals, and insurance companies make informed decisions.

3.1 Software Requirement Specification

Software Requirement Specification (SRS) describes the functional and non-functional requirements of the healthcare cost prediction system. It defines how the system should operate, the features it should provide, and the technologies required for implementation. The SRS helps developers understand system requirements clearly and ensures that the final system meets user expectations.

The healthcare cost prediction system is designed to analyze patient data and predict medical expenses using machine learning and explainable AI techniques. The system takes patient demographic and medical information as input, processes the data, and generates predicted healthcare costs. It also identifies important features influencing cost prediction and provides visualization for better understanding.

3.1.1 User requirement specifications

User requirements describe the needs and expectations of users who interact with the healthcare cost prediction system. The system is designed for patients, healthcare providers, hospitals, and insurance companies to predict healthcare costs accurately and support better decision-making. These users require a simple, reliable, and efficient system that can analyze healthcare data and generate cost predictions.

The users expect the system to allow easy data input, process healthcare information, and display predicted healthcare costs clearly. The system should also provide explanations for predictions so that users can understand which factors influence the cost. Additionally, users require graphical representations and reports for better analysis and interpretation of results.

- The user should be able to upload healthcare datasets
- The user should be able to enter patient details manually
- The system should predict healthcare costs accurately
- The system should display prediction results clearly
- The system should identify important factors influencing cost
- The user should be able to view graphs and visualizations
- The system should compare different machine learning models
- The system should generate reports for analysis
- The system should be easy to use and user-friendly
- The system should provide fast prediction results
- The system should handle large healthcare datasets
- The system should reduce manual work
- The system should provide explainable AI outputs
- The system should support decision-making for users

These user requirements ensure that the healthcare cost prediction system is useful, efficient, and easy to use for predicting healthcare expenses and improving healthcare management.

3.1.2 Software requirements

Software requirements specify the software tools, programming languages, libraries, and platforms needed to develop and implement the healthcare cost prediction system. These software components help in data processing, model training, prediction, and visualization of healthcare cost results.

The proposed system is developed using Python programming language because it provides powerful libraries for data analysis, machine learning, and explainable AI. The system uses data science tools and machine learning frameworks to build accurate healthcare cost prediction models and generate meaningful insights**.**

Software requirements define the technologies and tools used to build the system.

Frontend: React.js, HTML5, CSS3, JavaScript

The frontend is the user interface of the system.

- **React.js** is used to build interactive web pages
- **HTML5** structures the web page
- **CSS3** is used for styling and design
- **JavaScript** adds functionality and interactivity

These technologies help users easily input patient details, upload datasets, and view prediction results.

Backend: Python, FastAPI / Flask

The backend handles data processing and prediction logic.

- **Python** is used because it supports machine learning and data analysis

- **FastAPI / Flask** are web frameworks used to connect frontend and machine learning models.

The backend receives user data, processes it, runs the prediction model, and sends results back to the user interface.

Machine Learning Libraries: Scikit-learn, Pandas, NumPy, SHAP

These libraries are used for building and explaining the prediction model:

- Scikit-learn → Machine learning algorithms (HistGradientBoostingRegressor, Decision Tree, Lasso.)
- **Pandas** → Data processing and dataset handling
- **NumPy** → Mathematical and numerical operations
- **SHAP** → Explainable AI (shows which features affect cost prediction)

These libraries help improve prediction accuracy and transparency.

Database: MongoDB

- MongoDB is used to store patient data and prediction results
- It allows fast data storage and retrieval
- It is flexible and suitable for healthcare datasets

Tools: VS Code, Node.js, Python 3.10+, MongoDB Compass

These tools help in development:

- **VS Code** → Code editor
- **Node.js** → Required for React frontend
- **Python 3.10+** → Running machine learning models
- **MongoDB Compass** → Database management tool

3.1.3 Hardware requirements

Hardware requirements define the minimum system configuration needed to run the project.

Processor: Minimum Intel i3 or Equivalent

A processor is needed to run machine learning models and process data.
Intel i3 is sufficient for small to medium datasets.

RAM: At least 8GB (16GB Recommended)

- 8GB RAM → Minimum requirement.
- 16GB RAM → Recommended for faster machine learning training.

More RAM helps process large datasets efficiently.

Storage: 2–4 GB

Storage is needed for:

- Dataset files
- Machine learning models
- Required software and libraries

2–4 GB storage is enough for this project.

Table 3.1 hardware requirements

Hardware Component	**Requirement**	**Description**
Processor	Minimum Intel i3 or equivalent	Required to run machine learning algorithms and data processing tasks
RAM	At least 8 GB (16 GB recommended for ML training)	Helps in handling datasets and improves model training performance
Storage (Hard Disk / SSD)	2–4 GB	Required for storing datasets, trained models, and dependencies

3.2 EHCP Architecture

The content diagram shows the overall workflow of the Healthcare Cost Prediction system. First, healthcare data is collected from datasets containing patient information such as age, diagnosis, severity, and hospital details. The data is then preprocessed to remove missing values and clean the dataset. After preprocessing, feature selection is performed to identify the most important factors affecting healthcare costs.

Next, machine learning algorithms such as Random Forest, Decision Tree, and XGBoost are used to train the prediction model. The trained model is evaluated using performance metrics like accuracy and R^2 score. Explainable AI techniques such as SHAP are then applied to identify the important features influencing the cost prediction. Finally, the system generates predicted healthcare costs and displays results using graphs and reports.

figure 3.1 EHCP architecture.

The image represents a structured and disciplined system, symbolizing the organized workflow of the healthcare cost prediction model. Just like the uniform components work together efficiently, the project integrates data preprocessing, machine learning, and API services to deliver accurate predictions. It reflects reliability, coordination, and precision in system design and execution.

4. IMPLEMENTATION

4.1 Method of implementation

The Healthcare Cost Prediction System was implemented as a full-stack machine learning application that integrates data preprocessing, model training, evaluation, and deployment through a web-based dashboard for real-time predictions and analytics.

The system predicts hospital treatment costs using clinical and demographic attributes from open healthcare data.

Implementation Pipeline

1. Data Collection

System uses the New York State SPARCS inpatient discharge dataset, consisting of approximately 2.3 million de-identified patient records (2016).

The dataset includes:

- Length of Stay
- APR DRG Code
- APR Severity of Illness
- APR Risk of Mortality
- Emergency Department Indicator
- Payment Typology
- Gender
- Ethnicity
- Medical/Surgical Description
- Total Charges

2. Data Preprocessing

To prepare the dataset for model training, the following preprocessing steps were

performed:

- Removal of irrelevant or redundant columns
- Handling missing values
- Outlier clipping using statistical thresholds
- Log transformation applied to the target variable (Total Charges)
- Encoding categorical variables using **OneHotEncoder**
- Feature scaling for numerical attributes
- Train-test split (50% training, 50% holdout test set)
- This preprocessing ensured data consistency and improved model stability.

3. Feature Engineering

Feature engineering focused on selecting clinically relevant attributes that significantly influence healthcare costs.

Key features included:

- Length of Stay
- APR DRG Code
- APR Severity of Illness
- APR Risk of Mortality
- Emergency Department Indicator
- Payment Typology
- Demographic variables (Gender, Ethnicity)
- Feature importance analysis and SHAP explanations were used to validate feature contributions.

4. Model Training

The primary model used in this project is:

HistGradientBoostingRegressor

- This model was selected due to:
- High computational efficiency on large datasets
- Histogram-based binning for faster training

- Strong handling of non-linear medical cost relationships
- Better generalization performance
- Hyperparameters were optimized through experimentation. The target variable was trained on the **log-transformed cost values**, and predictions were converted back to the original scale for evaluation.

5. Model Evaluation

Model performance was evaluated using:

- R^2 Score (Coefficient of Determination)
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- 5-Fold Cross Validation Score

The trained model achieved:

- High R^2 value indicating strong predictive accuracy
- Low MAE indicating minimal average prediction error
- Stable cross-validation performance demonstrating generalization
- Performance metrics are displayed dynamically on the dashboard.

6. Backend Integration & Deployment

The trained model was integrated into a production-ready backend system:

- FastAPI → REST API development
- MongoDB → Patient data storage
- SHAP (SHapley Additive Explanations) → Model interpretability
- React.js Frontend → User dashboard and prediction interface

The backend provides:

- Real-time prediction endpoint
- Model metrics endpoint
- Patient record storage and retrieval
- Explainability outputs (feature contributions)

4.1.1 Technologies and libraries used

The system was implemented using Python-based machine learning and backend development frameworks. The major libraries and their roles in the project are summarized below:

1. FastAPI

Description:
FastAPI is used to develop REST APIs for real-time cost prediction and retrieval of model metrics. It allows the backend to communicate with frontend interfaces or external clients seamlessly.

Sample Usage:

```
from fastapi import FastAPI

from pydantic import BaseModel

app = FastAPI()

class PatientInput(BaseModel):

    length_of_stay: int

    apr_drg_code: int

@app.post("/predict")

def predict_cost(data: PatientInput):

    predicted_cost = 12345.67  # example prediction
```

```
    return {"predicted_cost": predicted_cost}
```

2. Pandas

Description:
Pandas is used for loading, cleaning, preprocessing, and feature engineering of the dataset.

Sample Usage:

```
import pandas as pd

# Load dataset

df = pd.read_csv("sparcs.csv")

# Data cleaning

df = df.dropna(subset=["Length of Stay", "Total Charges"])

# Feature engineering

df["log_los"] = (df["Length of Stay"] + 1).apply(np.log)
```

3. NumPy

Description:
NumPy provides support for numerical computations and mathematical transformations.

Sample Usage:

```
import numpy as np

los = np.array([1, 5, 10])

log_los = np.log1p(los)  # log(1 + LOS)
```

4. Scikit-learn

Description:
Scikit-learn is used for model training, preprocessing pipelines, cross-validation, and performance evaluation.

Sample Usage:

```
from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.ensemble import HistGradientBoostingRegressor

from sklearn.metrics import r2_score



X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.1,

random_state=42)

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)

X_test_scaled = scaler.transform(X_test)

model = HistGradientBoostingRegressor()

model.fit(X_train_scaled, y_train)

y_pred = model.predict(X_test_scaled)

print("R2 Score:", r2_score(y_test, y_pred))
```

5. HistGradientBoostingRegressor

Description:
This is the primary algorithm used for cost prediction. It is a tree-based boosting method that works efficiently with large datasets and provides high accuracy.

Sample Usage:

```
from sklearn.ensemble import HistGradientBoostingRegressor

hgb_model = HistGradientBoostingRegressor(max_iter=500, learning_rate=0.03)

hgb_model.fit(X_train, y_train)

y_pred = hgb_model.predict(X_test)
```

6. SHAP

Description:
SHAP is used for model interpretability, providing feature contribution analysis to understand how each feature influences predictions.

Sample Usage:

```
import shap

explainer = shap.TreeExplainer(hgb_model)

shap_values = explainer.shap_values(X_test)

shap.summary_plot(shap_values, X_test)
```

7. PyMongo

Description:
PyMongo is used to connect to MongoDB and manage patient records, allowing storage and retrieval of predictions and patient data.

Sample Usage:

```
from pymongo import MongoClient

client = MongoClient("mongodb://localhost:27017/")

db = client["medical_costs_db"]
```

```
collection = db["patients"]

collection.insert_one({"name": "John Doe", "predicted_cost": 12345.67})

patients = list(collection.find({}, {"_id": 0}))

print(patients)
```

8. Joblib

Description:
Joblib is used to save and load trained machine learning models efficiently for reuse in predictions.

Sample Usage:

```
import joblib

# Save model

joblib.dump(hgb_model, "best_model.joblib")

# Load model

loaded_model = joblib.load("best_model.joblib")

y_pred = loaded_model.predict(X_test)
```

4.1.2 EHCP Algorithm

HistGradientBoostingRegressor

The HistGradientBoostingRegressor is an optimized gradient boosting algorithm that builds an ensemble of decision trees sequentially.

Algorithm Steps

1. Data Preprocessing

- Load the SPARCS dataset (sparcs.csv).
- Handle missing values:
  - Numeric features (Length of Stay, APR DRG Code, APR MDC Code, APR Severity of Illness Code, Discharge Year) → imputed by removing rows with missing values.
  - Categorical features (Age Group, Gender, Race, Ethnicity, Type of Admission, Patient Disposition, APR Risk of Mortality, APR Medical Surgical Description, Payment Typology 1) → rows with missing values dropped.
- Remove outliers in Total Charges (top 1% capped).
- Encode categorical features using One-Hot Encoding.
- Scale numeric features using StandardScaler

2. Feature Engineering

- Engineered clinical and interaction features:

severity_x_mortality = APR Severity of Illness Code × APR Risk of Mortality Score

drg_x_severity = APR DRG Code × APR Severity of Illness Code

mdc_x_severity = APR MDC Code × APR Severity of Illness Code

log_los = log(Length of Stay + 1)

los_x_severity = Length of Stay × APR Severity of Illness Code

los_x_emergency = Length of Stay × Emergency Department Indicator

emergency_x_severity = Emergency Department Indicator × APR Severity of Illness Code

facility_code = numeric code derived from facility name

drg_frequency = number of times the same DRG appears in dataset

3. Model Initialization

```
from sklearn.ensemble import HistGradientBoostingRegressor

model = HistGradientBoostingRegressor(max_iter=500, learning_rate=0.03, max_depth=10, min_samples_leaf=5, l2_regularization=0.01, early_stopping="auto", validation_fraction=0.1, n_iter_no_change=20, random_state=42)
```

4. Model Training

- Train-test split: 90% training, 10% test.
- Fit the preprocessed training data into a **Pipeline** with preprocessor + model.
- Cross-validation (5-fold) used to select the best model among:
  - Decision Tree Regressor
  - HistGradientBoostingRegressor
  - Lasso Regression

5. Prediction

```
y_pred = model.predict(X_test)
```

6. Model Evaluation

- Metrics calculated:
  - R^2 score, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE)
- The selected model (HistGradientBoostingRegressor) achieved highest **CV R^2** and stable test performance.

7. Explainability using SHAP

```
import shap

explainer = shap.TreeExplainer(model.named_steps['estimator'])

shap_values = explainer.shap_values(X_test_transformed)
```

```
shap.summary_plot(shap_values, X_test_transformed)
```

- Provides **feature contribution insights** for each patient prediction.

Why This Algorithm Was Chosen

- Handles large-scale SPARCS dataset efficiently
- Provides better performance than traditional linear regression or decision trees
- Requires less training time compared to classical Gradient Boosting
- Captures non-linear relationships in healthcare cost data
- Supports interpretability via SHAP for actionable insights

4.2 Result analysis

The model achieved strong predictive performance with R^2 score above 0.91 on log-transformed data and approximately 0.88 on the original scale. The results indicate that the model explains more than 90% of the variance in healthcare costs. SHAP analysis confirmed that Length of Stay, Severity Level, and DRG Codes are major contributing factors.

5. TESTING

5.1 Features to be tested

- Input Validation and Error Handling

This feature ensures that the system correctly checks user input before processing. The model should accept only valid data such as age, gender, BMI, and medical details. If users enter incorrect or missing values, the system should display proper error messages and guide users to provide valid inputs. This helps prevent incorrect predictions and improves system reliability.

- Model Prediction Accuracy

This feature tests how accurately the machine learning model predicts healthcare costs. The predicted values are compared with actual costs using evaluation metrics such as R^2 score, Mean Absolute Error (MAE), or Root Mean Square Error (RMSE). High accuracy indicates that the model is performing well and producing reliable predictions.

- SHAP Explanation Generation

SHAP (SHapley Additive exPlanations) is used to explain model predictions. This feature tests whether the system correctly generates explanations showing how each input feature (like age, BMI, or disease severity) contributes to the predicted healthcare cost. This improves transparency and helps users understand the prediction results.

- MongoDB Data Storage and Retrieval

This feature ensures that the system properly stores input data, prediction results, and explanations in MongoDB. It also checks whether stored data can be retrieved quickly and accurately when required. This helps in maintaining records and improving system performance.

- API Response Correctness

The system uses APIs to communicate between frontend and backend. This feature tests whether APIs return correct prediction results, status messages, and error responses. It ensures smooth communication and correct data transfer between system components.

- Frontend-Backend Integration

This feature checks whether the user interface properly interacts with the backend model. When users enter data in the frontend, it should be sent to the backend, processed by the model, and results should be displayed correctly. Proper integration ensures smooth user experience and system functionality.

5.2 Testing Tools and Environment

The testing of the Healthcare Cost Prediction system is carried out using various tools and technologies to ensure proper functioning, accuracy, and performance of the system.

- Windows OS

Windows Operating System is used as the development and testing environment. It provides a stable platform for running Python, backend services, and frontend applications. All system components are developed and tested on Windows for compatibility and smooth execution.

- Python 3.x

Python 3.x is used as the main programming language for building and testing the machine learning model. It is used for data preprocessing, model training, prediction, and integration with backend services. Python also supports libraries required for explainable AI and machine learning.

- FastAPI

FastAPI is used to build backend APIs. It helps in testing how the frontend communicates with the machine learning model. FastAPI also enables testing of API responses, input validation, and prediction output delivery.

- MongoDB

MongoDB is used as the database for storing prediction results, user inputs, and model explanations. Testing ensures that data is properly stored, retrieved, and managed within the database.

- Scikit-learn

Scikit-learn is used for implementing machine learning algorithms such as regression and ensemble models. Testing is performed to ensure the model is trained correctly and produces accurate healthcare cost predictions.

- SHAP

SHAP is used for explainable AI. It helps test whether the system correctly generates explanations for predictions. Testing ensures that feature importance and contribution values are displayed properly.

- React.js

React.js is used to develop the frontend user interface. Testing ensures that user inputs are collected correctly and prediction results are displayed properly. It also checks frontend and backend integration.

5.3 Design of test cases and scenarios

- Database and Data Integrity Testing

Data persistence is fundamental in healthcare-based systems, as patient-related information must be stored and retrieved accurately without corruption. The MongoDB database layer was therefore subjected to extensive validation to guarantee operational integrity. Testing procedures confirmed accurate insertion of patient records, ensuring that all submitted attributes are stored with correct schema alignment and without truncation or structural modification.

Retrieval operations were verified to ensure that stored records are returned exactly as inserted, preserving numerical precision and categorical values. Deletion

operations were also evaluated to confirm that specific records can be removed without affecting unrelated entries, thus preventing unintended data loss. Additional validation measures were implemented to detect potential duplication of records, enforce unique identifiers where necessary, and prevent data inconsistencies during concurrent write operations. Through these tests, the persistence layer demonstrated high reliability and consistency, ensuring long-term data stability and operational trustworthiness.

- Integration and End-to-End Workflow Testing

While individual module validation is important, true system reliability can only be verified through comprehensive end-to-end testing. The entire workflow—from frontend interaction to final output rendering—was rigorously evaluated to ensure seamless integration across all components. The validated processing pipeline follows the sequence:

Frontend → API → Preprocessing Pipeline → Machine Learning Model → SHAP Explainability → Database Storage → Response Rendering

Each stage was monitored to ensure consistent data exchange and format compatibility. Special attention was given to verifying that preprocessing transformations applied during model training are identically executed during inference, thereby eliminating feature misalignment issues. The consistency of SHAP feature contributions was also evaluated to confirm that explanations correspond to correctly transformed features.

Response rendering at the frontend level was validated to ensure accurate visualization of predictions and feature contribution outputs. Integration testing confirmed that no transformation mismatches, serialization errors, or structural inconsistencies occurred during data transfer between layers. This comprehensive validation guarantees that the system operates as a cohesive and synchronized architecture rather than isolated modules.

- Performance and Scalability Evaluation

Given the real-time nature of predictive healthcare applications, system performance and scalability were systematically assessed. Performance testing focused on three primary metrics: prediction latency, SHAP computation time, and overall API response duration. Prediction latency was measured from the moment a request was received to

the generation of the regression output. SHAP computation time was separately analyzed due to its computational complexity, particularly for tree-based ensemble models.

The total API response time, encompassing preprocessing, model inference, SHAP value generation, database interaction, and JSON response formatting, was maintained below the defined threshold of three seconds per request. This benchmark ensures practical usability in real-world healthcare environments.

To assess scalability, concurrent request simulations were performed to evaluate system stability under multiple simultaneous users. The system demonstrated stable memory utilization and consistent response times without degradation. Monitoring confirmed the absence of memory leaks, thread deadlocks, or computational bottlenecks during extended testing cycles. These results indicate that the system is capable of maintaining efficiency and reliability under realistic operational workloads.

- Reliability and Robustness Testing

Reliability testing was conducted to evaluate long-term operational stability under continuous usage conditions. The system was subjected to repeated execution cycles to simulate prolonged deployment scenarios. These tests verified that repeated predictions and SHAP computations do not accumulate hidden state errors or degrade performance over time.

Negative testing strategies were also implemented to intentionally provide invalid, incomplete, or extreme input values. The objective was to ensure that the system handles such scenarios gracefully without producing undefined outputs or runtime exceptions. Stress testing under high-load conditions further validated the structural resilience of the backend infrastructure. Even under increased request frequency, the system maintained controlled response behavior and preserved data integrity. Collectively, these tests confirm that the application remains stable, secure, and dependable under both expected and adverse operating conditions.

- Acceptance Criteria

The system validation framework defines clear acceptance criteria to determine readiness for deployment. The application is considered validated when statistical model performance meets predefined thresholds, including satisfactory R^2 and error

metrics on validation datasets. Furthermore, SHAP explainability must satisfy its theoretical additivity property across all evaluated samples, ensuring mathematical correctness of feature contribution decomposition.

Feature alignment between preprocessing transformations and SHAP outputs must remain consistent to preserve interpretability. Explanations must also align with domain-level healthcare logic, reinforcing trust in predictive reasoning. All functional and integration test cases must pass successfully without critical failures. Additionally, performance benchmarks, including response time constraints, must be satisfied. Finally, the absence of unhandled runtime exceptions across testing scenarios serves as a key indicator of production-level robustness.

5.5 User Interface of EHCP

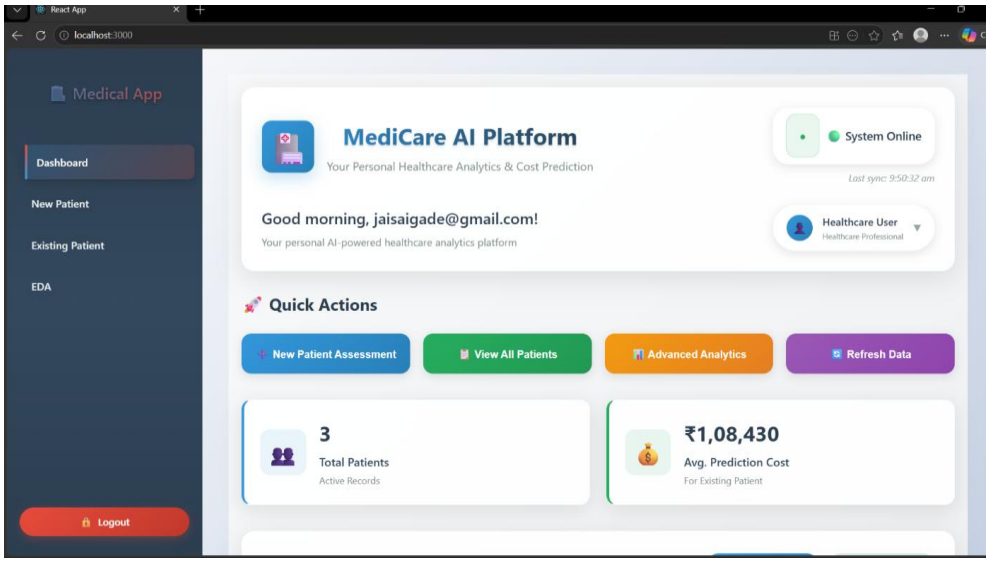

figure 5.1 medicare ai platform dashboard page

The Medicare AI Platform dashboard is the main page of the healthcare cost prediction system. It provides an overview of the system and allows users to manage patient data easily. On the left side, there is a navigation menu with options such as Dashboard, New Patient, Existing Patient, EDA, and Logout. The top section displays the system name, user information, and system status showing that the platform is online. The Quick Actions section contains buttons like New Patient Assessment, View All Patients, Advanced Analytics, and Refresh Data, which help users perform tasks quickly. The dashboard also shows summary information such as the total number of patients and the average predicted treatment cost. Overall, this dashboard helps healthcare

professionals add patient details, predict treatment costs, view records, and analyze healthcare data in one place.

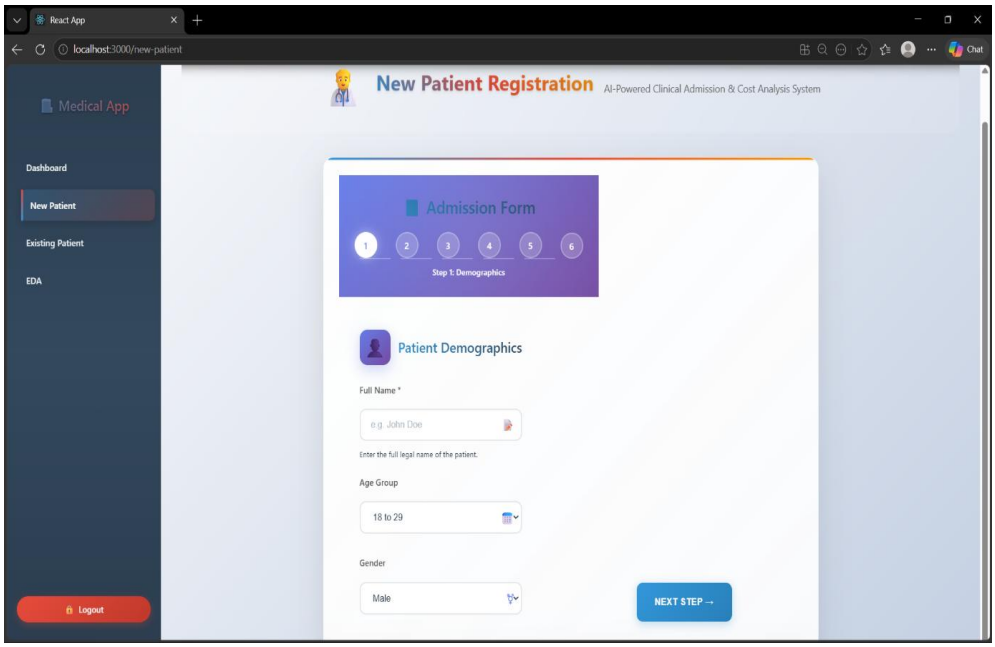

figure 5.2 new patient registration page

This screen shows the New Patient Registration page of the Medicare AI Platform. It is used to add details of a new patient into the system for healthcare cost prediction. On the left side, there is a navigation menu with options like Dashboard, New Patient, Existing Patient, EDA, and Logout. In the main section, there is an Admission Form divided into multiple steps, and currently, Step 1: Demographics is displayed. This section collects basic patient information such as Full Name, Age Group, and Gender. After entering the required details, the user can click the Next Step button to move to the next stage of the form. Overall, this page helps healthcare staff register a new patient by entering their basic details before predicting treatment costs.

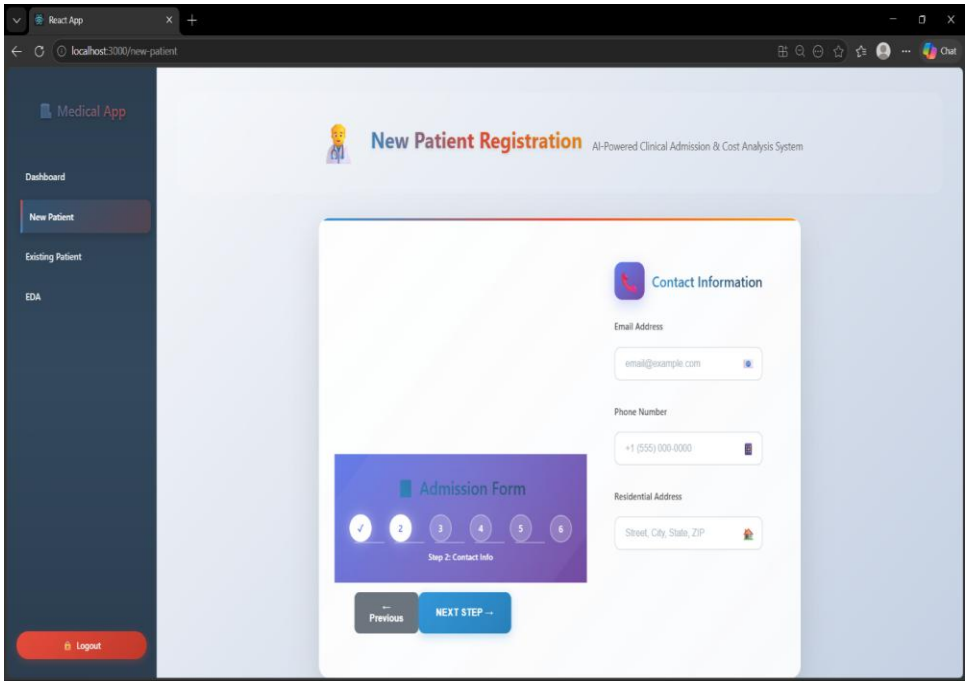

figure 5.3 patient contact information

This screen shows Step 2: Contact Information in the New Patient Registration process of the Medicare AI Platform. In this step, the user enters the patient's contact details such as Email Address, Phone Number, and Residential Address. These details help the hospital maintain communication and store patient information properly. The form also shows the progress bar indicating that this is the second step out of multiple steps in the admission form. At the bottom, there are Previous and Next Step buttons, allowing the user to move back to the previous section or continue to the next step. Overall, this page is used to collect patient contact details before moving forward in the registration process.

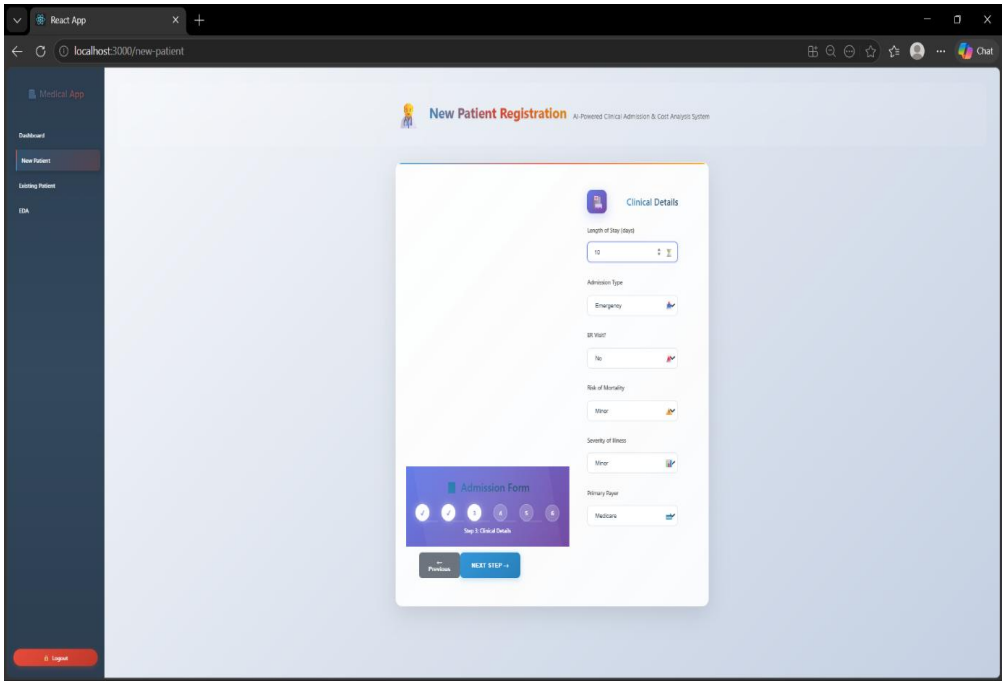

figure 5.4 clinical details

This screen shows Step 3: Clinical Details in the New Patient Registration process of the Medicare AI Platform. In this step, the user enters important medical information about the patient such as Length of Stay, Admission Type, ICU Visit, Risk of Mortality, Severity of Illness, and Billing Source. These clinical details help the system understand the patient's medical condition and are used to predict the healthcare treatment cost accurately. The progress bar at the bottom shows that this is the third step in the admission form. The Previous button allows the user to go back to the earlier step, while the Next Step button moves to the next stage. Overall, this page collects medical details required for accurate healthcare cost prediction.

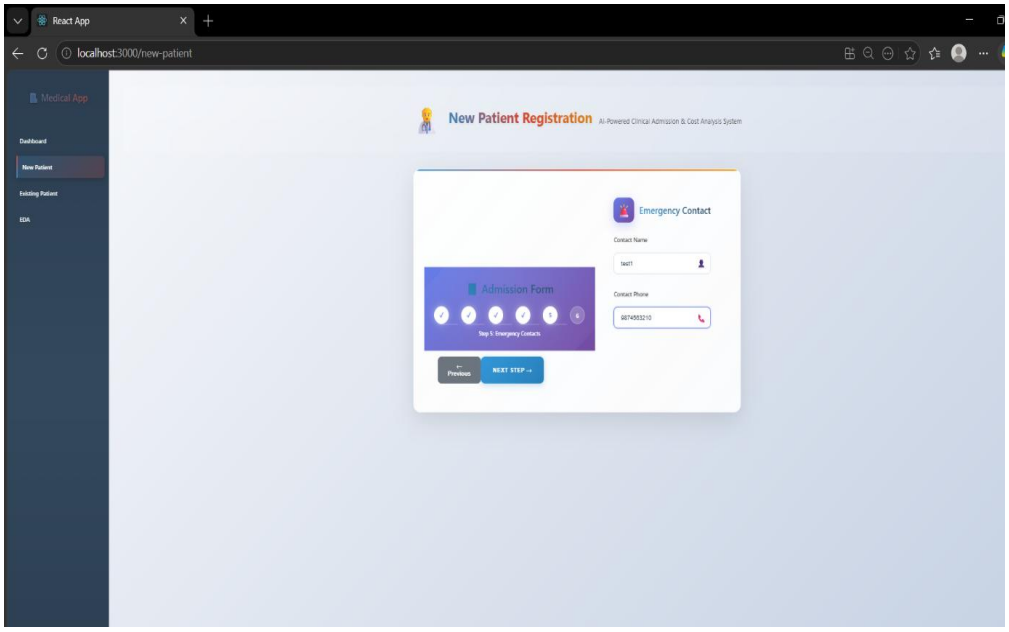

figure 5.5 emergency contact

This screen is part of a hospital or clinic's software used to register new patients. The form shown is focused on collecting emergency contact details, such as the contact person's name and phone number. On the left side, there is a navigation menu with options like Dashboard, Patient Form, and Setting Patient, which help staff move between different sections of the system. At the bottom of the form, a NEXT STEP button indicates that this is only one stage of a larger, step-by-step admission process. The circles above the button act as a progress tracker, showing how far along the registration is. The web address displayed suggests that the application is currently being tested locally on a computer rather than being accessed through a public website. In simple terms, this is a patient registration form that guides hospital staff through entering information step by step, starting with emergency contact details.

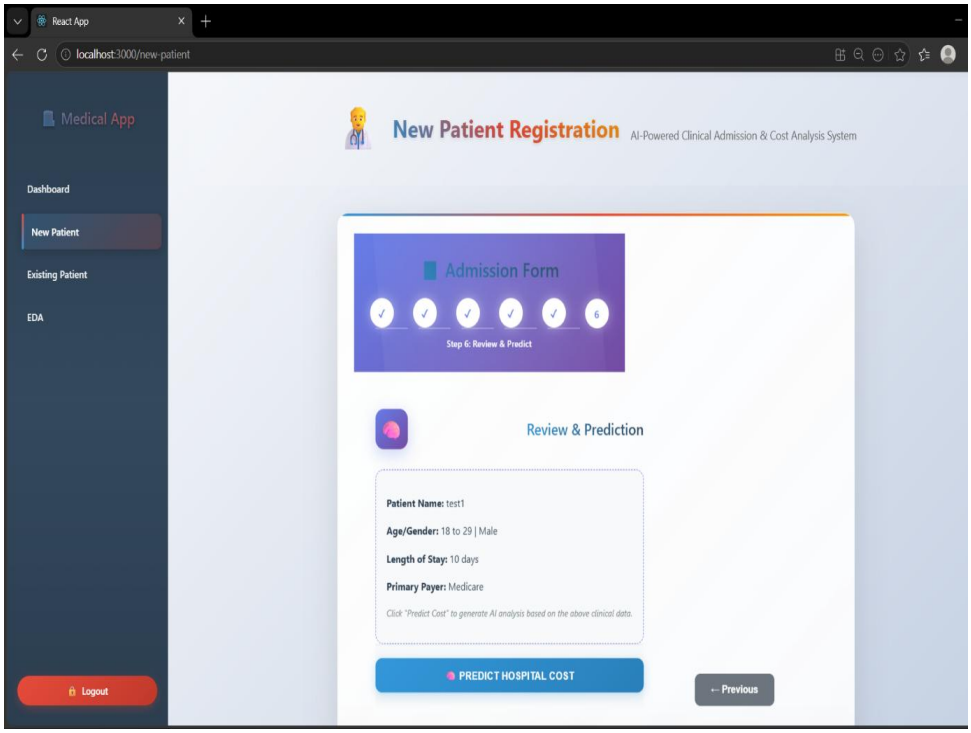

figure 5.6 review & prediction

This screen belongs to a medical app that helps hospitals register patients and also use AI to estimate treatment costs. The section shown is called “Step 3: Review & Predict”, which means the staff have already entered patient details in earlier steps and now they are reviewing the information. The form displays basic details like the patient’s name, age, gender, how long they will stay in the hospital, and their insurance provider. There is a button 40abelled “Predict Hospital Cost”, which uses AI to calculate the expected cost of the patient’s stay based on the data provided. In simple words, this part of the system lets hospital staff check the patient’s information and then click a button to get an AI-generated prediction of how much the hospital stay will cost.

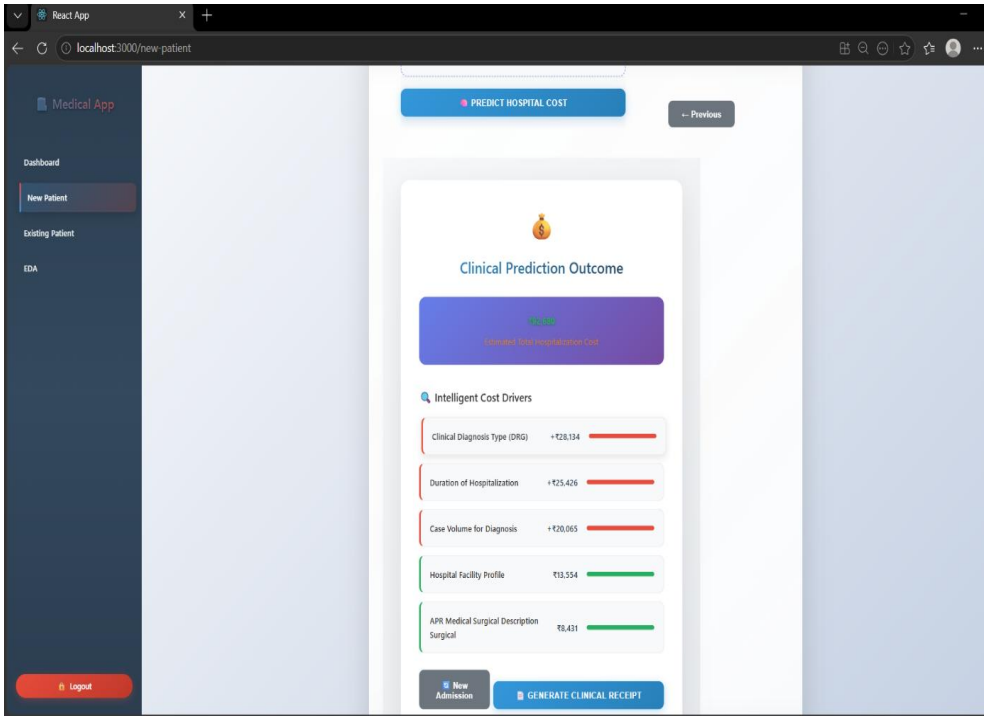

figure 5.7 cost prediction

This screen is part of a medical app that shows the results of an AI prediction about hospital costs. At the top, it displays a prediction score which represents the outcome of the analysis. Below that, there's a section called "Intelligent Cost Drivers" listing the main factors that influenced the prediction. These include the type of clinical diagnosis, how long the patient stays in the hospital, the number of similar cases, the quality of the hospital facilities, and a specific medical description. Each factor has a percentage bar showing how much it contributes to the overall prediction. At the bottom, there are two buttons: one to start a new admission and another to generate a clinical receipt. In simple words, this screen explains why the AI predicted a certain hospital cost by showing the key reasons behind it.

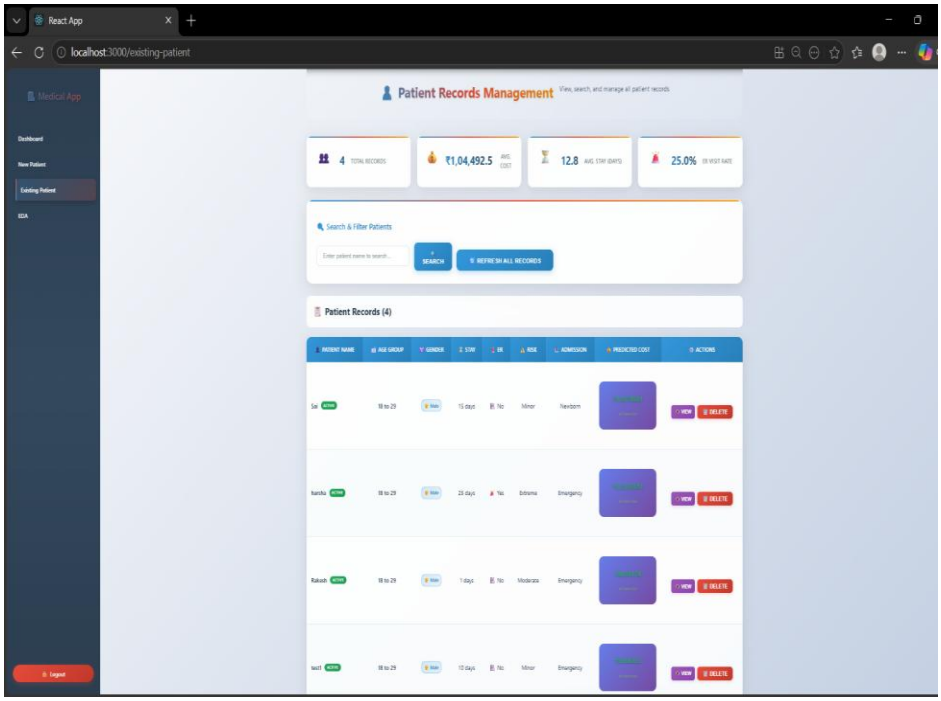

figure 5.8 patient record management

This screen is part of a healthcare management system where patient records are organized and displayed in a dashboard. At the top, it shows summary statistics such as the total number of patients, their average age, and the overall recovery rate. Below that, there's a search bar that allows staff to quickly find or filter patient information, along with a button to add new patients. The main section lists patient details in a table format, including ID, name, age, gender, diagnosis, and current status, with options to edit or delete each record. On the left side, a sidebar provides navigation to different sections like Dashboard, Patient Records, and Settings, with a logout button at the bottom. In simple words, this screen is a digital tool that helps hospitals or clinics keep track of patient information, search for specific records, and update or manage them easily.

6. CONCLUSION & FUTURE ENHANCEMENT

This project successfully developed a healthcare cost prediction system using machine learning. The model predicts hospital treatment costs based on factors such as length of stay, admission type, risk severity, emergency cases, and patient demographics. Proper data preprocessing, feature engineering, and model training helped improve prediction accuracy.

A unified training and prediction pipeline was created to ensure consistent and reliable results. The model shows logical behavior, where higher risk and longer hospital stays result in higher predicted costs. This confirms that the system learned meaningful healthcare patterns.

The integration of SHAP improves transparency by explaining how each feature affects the predicted cost. Additionally, the dashboard provides useful insights like average cost and emergency rate, helping in better hospital planning.

Overall, the project successfully demonstrates a practical, accurate, and explainable healthcare cost prediction system, with future scope for cloud deployment, advanced models, and real-time data integration.

The Healthcare Cost Prediction system can be further enhanced by integrating real hospital datasets such as SPARCS to improve prediction accuracy and make the system more practical for real-world applications. Cloud deployment can be implemented to improve scalability, accessibility, and system performance. Advanced machine learning models such as XGBoost and Deep Learning can be used to enhance prediction accuracy and model efficiency. The system can also be extended to include insurance fraud detection to identify unusual claims and reduce financial risks. Additionally, risk prediction and preventive analytics can be incorporated to help healthcare providers identify high-risk patients and take preventive measures.

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